

Product Roadmap and SW Architecture

****Note:** any changes to the roadmap will be documented formally.

Product Roadmap

S1.R0 — Planning & Requirements (Week 1–2)

Milestones

- Define personas (Counselor, Office Admin, Practice Owner)
- Define primary workflows (schedule → session → notes → plan updates)

Deliverables

- User stories + acceptance criteria
- Architecture draft

S2.R1 — MVP (Core Workflow) (Week 3–5)

Goal: Make the bare-bones for a counselor: client + appointments + basic notes.

Key features

- Client profiles (filters)
- Session note template (simple SOAP or freeform)
- Contains a single text body. (for structural purposes between FE and BE)
- Is linked to a specific appointment.
- Is editable after creation.
- Persists in the database.

Success criteria

- A counselor can actively use the software to assist them in diagnoses and treatment plans during any given session.

S2.R3 — Treatment Plan Builder (Week 3–5)

Goal: Reduce time spent creating treatment programs & finding the treatment plan resources.

Key features

- Treatment plan templates (e.g., Anxiety, Depression, Substance Use) with editable goals/objectives

Each treatment plan will contain:

- A defined problem area (Anxiety, depression, procrastination, etc.)
- Goals for improvement
- Goals should have metrics for improvement (number of tasks done, time spent studying, grounding methods)

Counselor must be able to

- Add, edit, and remove goals
- Add, edit, and remove objectives within each goal
- Save plans as draft or finalized
- Retrieve and modify previously created plans

Plans will be stored in a structured JSON format within the TreatmentPlans table and linked to a specific client.

Export to PDF for records/share

Success criteria

- Draft treatment plan created in <10 minutes.

S3.R4 — Automation & Practice Operations (Week 6–8)

Goal: Automate repetitive work + add practice-level value.

Key features

- Intake questionnaire (client-facing form)
- Role support: Admin vs Counselor permissions (optional)

Success criteria

- Counselor administrative tasks greatly reduced.

S4.R5 — Hardening, Compliance, and Launch (Week 9–11)

Goal: Make it stable, secure, and demo-ready.

Key features

- Audit log (who accessed/changed client data)
- Backups + restore plan
- Performance fixes + usability polish
- Deployment (cloud) + final documentation

Success criteria

- Demo runs reliably; security basics are in place; minimal bugs.

SW Architecture

1) Database

Recommended: PostgreSQL (treatment plan data)

- **Core tables/entities**
 - SessionNotes (appointment_id, content, created_at)
 - TreatmentPlans (client_id, plan_json, version, status)
 - AuditLogs (user_id, action, entity_type, entity_id, timestamp)
- **Why relational?**
 - Relationships (clients ↔ appointments ↔ notes) are clean in SQL.
- **Encryption**
 - Encrypt sensitive fields at rest where possible; always TLS in transit.

2) Platform

Recommended platform: Web application (responsive) + optional mobile-friendly UI

- **Frontend:** PHP
- **Backend:** SQL + Python
- **Why web-first?**
 - Easier deployment, cross-device access for clinics, fast iteration.

3) Server (Application + Hosting)

Architecture style: “Modular monolith” for MVP → event-driven components later

- **MVP:** single backend service + database
- **Why not microservices initially?**
 - Faster for a class project; less DevOps overhead; still scalable for small practices.
- **Deploy options**
 - Simple: Render / Fly.io / Azure App Service
 - Azure if we can confirm free tier provides necessary services
 - Database: Managed Postgres
- **Key server components**
 - API layer (REST endpoints)
 - Business logic modules (Notes, TreatmentPlans)
 - Background jobs (reminders, exports) via a worker process

4) Open-source / Third-party Tools

Use open-source strategically to reduce build time:

- **Auth/security:** bcrypt/argon2 for passwords; firebase/JWT library

- **UI components:** component library (MUI / shadcn / Bootstrap)
- **PDF export:** server-side PDF generator library
- **Calendar UI:** FullCalendar, Google Calendar API, or similar
- **AI-assist (optional)**
 - Keep it as a draft generator only; clinician reviews before saving.
 - Store prompts/outputs carefully; avoid sending PHI to external services unless explicitly permitted by your assignment constraints.

5) Development Tools

- **Source control:** Git + GitHub
- **Project management:** using Teampate, Kanban board (Backlog → Planned → In Progress → Done)
- **CI/CD:** GitHub Actions (run tests/lint on push)
- **Containerization:** Docker, or something similar
- **Testing**
 - Unit tests for scheduling + plan generation logic
 - API tests for critical endpoints (create client, create appointment, add note)
- **Documentation**
 - README with setup steps
 - API endpoint list
 - Architecture diagram + data model diagram